

The Two-Player $3n \pm 1$ Game

An Audited Binary Core, Strengthened Local Normalizations,
and the Exact Remaining Global Barrier

Version 7.0

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<https://github.com/Grisha-Pochuev/3n-plus-minus-1-game>

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Audit status of Version 7.0. This document does *not* claim a proof of the all-start no-draw conjecture. Earlier versions did make that claim, but hostile review found that their global marked-router argument used finite states as ranked proof tokens without first proving the required provenance, and compressed an attached short-tail module into an unproved exhaustiveness assertion. Those claims are withdrawn here. Every unconditional theorem in this version is proved in the text. The global conclusion is stated only as a conditional theorem under four explicit open obligations. Thus Version 7.0 is intended to be gap-free as a status manuscript, not to conceal an open step inside a claimed prize solution.

Abstract

The two-player $3n \pm 1$ game is played on positive odd integers. A move chooses $3n - 1$ or $3n + 1$, removes all powers of two, and wins immediately when the result is 1. Arbitrary play can cycle, so the question is whether optimal play nevertheless has finite remoteness from every start. We retain the exact binary conjugation used in earlier manuscripts, in which the two moves are an expanding map A and a strictly contracting map B . We then separate four notions that earlier drafts sometimes conflated: a finite outcome, a routing witness, a retained proof token, and a temporary arithmetic cursor.

Two local repairs are proved completely. First, a reverse fixed-tail step from $Q_r^e(3c)$ either removes one factor of three while preserving an actual draw, or exposes a genuine loss state and a strictly lower win child before the draw continues. Second, the exponent-one predecessor has the same complete dichotomy; in the exceptional residue classes its two continuing children form an exact adjacent lift over the actual loss sibling. We also give a corrected audit of the source-zero valuation-one row and of the long high-return pair. These results close several former local gaps, but they do not establish the global provenance invariant required to iterate the marked router indefinitely. The remaining obligations are isolated precisely, and a complete order-theoretic proof shows that they would imply the no-draw conjecture. A companion Python program checks the arithmetic identities and permanent counterexamples used in the audit; it is explicitly supporting evidence rather than a proof of the infinite theorem.

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1 The game, outcomes, and the exact status

The current state is a positive odd integer n . From $n > 1$ the player to move chooses one of

$$3n - 1, \quad 3n + 1,$$

and repeatedly divides the chosen integer by 2 until it is odd. Reaching 1 wins immediately. Equivalently, state 1 is terminal and losing for the player to move.

For an infinite game graph, the finite-remoteness outcome classes are defined inductively. A state is WIN when it has a LOSS child; it is LOSS when all of its children are WIN. These are the least classes closed under the two rules. Every remaining state is called DRAW.

Lemma 1.1 (Elementary draw recursion). *A DRAW state has no LOSS child and has at least one DRAW child.*

Proof. A state with a LOSS child is WIN. A state all of whose children are WIN is LOSS. Therefore a state in neither finite class has no LOSS child and cannot have all children WIN; at least one child is DRAW. $\square$

Arbitrary play does not terminate: in the original game, suitable choices give $5 \rightarrow 7 \rightarrow 5$. The target is therefore not termination under all choices.

Conjecture 1.2 (No-draw conjecture). Every positive odd starting state has finite remoteness. Equivalently, the game has no DRAW positions and optimal play reaches 1 after finitely many moves.

Conjecture 1.2 is the target of the EUR 500 problem posed by Ingo Althoefer [4]. It is *not* promoted to a theorem in this manuscript.

2 Binary conjugation

Write an original odd state as $n = 2m + 1$ and define

$$F(m) = \left\lceil \frac{3m}{2} \right\rceil = \frac{3m + (m \bmod 2)}{2}.$$

For later factorization, also put

$$J(s) = 2F(s) + 1 = \begin{cases} 3s + 1, & s \text{ even,} \\ 3s + 2, & s \text{ odd.} \end{cases}$$

For a positive integer x , let $R(x)$ be the integer obtained by deleting the maximal alternating suffix of the binary expansion of x , and let $\text{asl}(x)$ denote the number of deleted bits. Put $R(0) = 0$ and $\text{asl}(0) = 0$. For example,

$$R(1001_2) = 10_2, \quad R(10101_2) = 0.$$

Lemma 2.1 (Alternating-suffix factorization). *For every positive integer z , put $\delta = 1 - (z \bmod 2)$. There is an integer $k \geq 1$ such that*

$$3z + 1 + \delta = 2^k J(R(z)).$$

If $R(z) > 0$, k is the length of the maximal alternating suffix of z ; if $R(z) = 0$, k is one more than the binary length of z .

Proof. First suppose $r = R(z) > 0$, let $k = \text{asl}(z)$, and put $\epsilon = r \bmod 2$. The deleted alternating word has the exact numerical value

$$t_{k,0} = \left\lfloor \frac{2^k}{3} \right\rfloor, \quad t_{k,1} = \left\lfloor \frac{2^{k+1}}{3} \right\rfloor,$$

according as $\epsilon = 0$ or 1 , so $z = 2^k r + t_{k,\epsilon}$. Since $J(r) = 3r + 1 + \epsilon$, direct substitution in the two parity cases gives

$$3t_{k,\epsilon} + 1 + \delta = 2^k (1 + \epsilon).$$

Adding the contribution $3 \cdot 2^k r$ yields the claimed identity. If $R(z) = 0$, the whole binary word is alternating and starts with 1. Let ℓ be its binary length. Then $z = t_{\ell,1}$, and the same calculation gives

$$3z + 1 + \delta = 2^{\ell+1} = 2^{\ell+1} J(0).$$

Thus the exponent in the statement is $k = \ell + 1$. □

Proposition 2.2 (First normal form). *For $m > 0$, the two children in m -coordinates are exactly*

$$F(m), \quad F(R(m)).$$

Proof. Since

$$3(2m + 1) - 1 = 2(3m + 1), \quad 3(2m + 1) + 1 = 2(3m + 2),$$

exactly one of $3m + 1$ and $3m + 2$ is odd. If m is even, the direct odd expression is $3m + 1$ and its new m -coordinate is $3m/2 = F(m)$. If m is odd, the direct odd expression is $3m + 2$ and its new coordinate is $(3m + 1)/2 = F(m)$.

The other expression is

$$3m + 2 - (m \bmod 2) = 3m + 1 + \delta, \quad \delta = 1 - (m \bmod 2).$$

Lemma 2.1 says that its odd part is $J(R(m))$. Returning to m -coordinates gives

$$\frac{J(R(m)) - 1}{2} = F(R(m)).$$

Thus the two children are exactly the displayed pair. $\square$

Every child in Proposition 2.2 lies in the image of F . Representing the state $F(q)$ by q gives the conjugated game

$$A(q) = F(q) = \left\lceil \frac{3q}{2} \right\rceil, \quad B(q) = R(F(q)),$$

with terminal state $q = 0$.

Proposition 2.3 (Strict contraction). *For every $q > 0$,*

$$B(q) < q.$$

Proof. The deletion defining R removes at least one binary digit, so

$$B(q) \leq \left\lfloor \frac{F(q)}{2} \right\rfloor \leq \left\lfloor \frac{3q+1}{4} \right\rfloor < q$$

for $q \geq 2$. Directly, $B(1) = 0$. $\square$

After one original move, the m -coordinate is in the image of F . Consequently a proof that every conjugated state q has finite remoteness would imply Conjecture 1.2; the first original move would then lead to one of two finite-remoteness states.

3 Constant tails and coefficient sources

For a positive odd integer a , a phase $e \in \{0, 1\}$, and $r \geq 1$, define

$$Q_r^e(a) = a2^r - e.$$

This is the unique representation of a positive binary word whose maximal constant suffix consists of r copies of the bit e .

Lemma 3.1 (Long-tail recurrence and shared boundary). *For every $r \geq 3$,*

$$A(Q_r^e(a)) = Q_{r-1}^e(3a), \quad B(Q_r^e(a)) = Q_{r-2}^e(3a).$$

Moreover,

$$B(Q_1^e(a)) = B(Q_2^e(a)).$$

Proof. The formula for A follows directly from the two phase cases:

$$A(a2^r) = 3a2^{r-1}, \quad A(a2^r - 1) = 3a2^{r-1} - 1.$$

When $r \geq 3$, the image still ends in at least two equal bits. Its maximal alternating suffix therefore consists only of the last bit, and applying R removes one further constant bit, proving the formula for B .

At the boundary,

$$A(Q_1^e(a)) = 3a - e, \quad A(Q_2^e(a)) = 6a - e = 2(3a - e) + e.$$

The last bit of $3a - e$ is $1 - e$, so the appended bit e extends the same maximal alternating suffix by one digit. Deleting it leaves the same prefix in both cases. $\square$

Recall that $J(s) = 2A(s) + 1$. The map J is an increasing bijection from $\mathbb{N}_0$ to the positive odd integers not divisible by 3: explicitly, $6u + 1 = J(2u)$ and $6u + 5 = J(2u + 1)$. Thus every positive odd coefficient has a unique factorization

$$a = 3^k J(s), \quad k \geq 0.$$

We call s its *coefficient source*. Every positive state has unique canonical coordinates

$$q = Q_r^e(3^k J(s)).$$

Indeed, if q is even, take $e = 0$, $r = v_2(q)$, and $a = q/2^r$; if q is odd, take $e = 1$, $r = v_2(q + 1)$, and $a = (q + 1)/2^r$. In both cases a is positive and odd, and its unique factorization above determines k and s . We write $\rho(q) = s$ for this state source.

Lemma 3.2 (Ordinary side relation). *Let $q > 0$. If*

$$q \not\equiv 1, 3, 12, 14 \pmod{16},$$

then

$$B(A(q)) \in \{A(B(q)), B(B(q))\}.$$

If q is in one of the four exceptional classes, then $A(q)$ is not in an exceptional class.

Proof. Put $x = A(q)$, $r = R(x) = B(q)$, and $k = \text{asl}(x)$. In the nonexceptional classes one has $r > 0$: if $r = 0$, then x is wholly alternating, and the two short words are $10_2, 101_2$ while every longer word ends in 0101 or 1010 . Directly, the short words come from $q = 1, 3$, and

$$\begin{aligned} A(q) \equiv 10 \pmod{16} &\implies q \equiv 1, 12 \pmod{16}, \\ A(q) \equiv 5 \pmod{16} &\implies q \equiv 3, 14 \pmod{16}. \end{aligned}$$

Thus $r = 0$ places q in an exceptional class. A direct residue check gives

k	$q \pmod{16}$
1	0, 2, 5, 7, 8, 10, 13, 15
2	4, 6, 9, 11.

Thus every nonexceptional residue has $k \in \{1, 2\}$. We now treat these two actual suffix lengths, so no fixed modulus is being used to decide an unbounded suffix.

When $k = 1$, the deleted bit equals the last bit $\epsilon = r \bmod 2$, and $x = 2r + \epsilon$. Directly,

$$A(x) = 2A(r) + \epsilon.$$

Thus $A(x)$ is obtained from $A(r)$ by appending the bit ϵ . If this bit equals the last bit of $A(r)$, the maximal alternating suffix stops at the boundary and $R(A(x)) = A(r)$; otherwise it continues through exactly the maximal alternating suffix of $A(r)$ and $R(A(x)) = R(A(r)) = B(r)$.

When $k = 2$, the deleted word is 01 if r is even and 10 if r is odd. Direct substitution gives

$$A(x) = \begin{cases} 4A(r) + 2, & r \text{ even,} \\ 4A(r) + 1, & r \text{ odd.} \end{cases}$$

Hence $A(x)$ is obtained from $A(r)$ by appending respectively 10 or 01 . The same boundary test leaves either $A(r)$ or $R(A(r)) = B(r)$. Since $R(A(x)) = B(A(q))$, the first assertion follows.

For completeness, if $q = 16u + c$ with $c \in \{1, 3, 12, 14\}$, then

$$A(q) \bmod 16 \in \begin{cases} \{2, 10\}, & c \in \{1, 12\}, \\ \{5, 13\}, & c \in \{3, 14\}. \end{cases}$$

None of these residues is exceptional. □

Lemma 3.3 (Source-boundary selector). *For $s > 0$, define*

$$\alpha(s) = 1 - (\lfloor s/2 \rfloor \bmod 2).$$

Factor

$$3J(s) + 1 - 2e = 2^v J(t), \quad e \in \{0, 1\}.$$

Then

$$(v, t) = \begin{cases} (1, A(s)), & e = \alpha(s), \\ (v, B(s)) \text{ with } v \geq 2, & e = 1 - \alpha(s). \end{cases}$$

Proof. Put $x = A(s)$. Its parity is $1 - \alpha(s)$. Since $J(s) = 2x + 1$,

$$3J(s) + 1 - 2e = 2(3x + 2 - e).$$

For $e = \alpha(s)$ the quantity in parentheses is odd and equals $J(x)$: it is $3x + 1$ when x is even and $3x + 2$ when x is odd. This gives $v = 1$ and $t = A(s)$.

For the opposite phase $e = x \bmod 2$, the quantity in parentheses is

$$3x + 2 - (x \bmod 2) = 3x + 1 + (1 - (x \bmod 2)).$$

Lemma 2.1 factors it as a positive power of two times $J(R(x)) = J(B(s))$. Including the leading factor two gives $v \geq 2$ and $t = B(s)$. $\square$

Lemma 3.4 (Exceptional child-pair identity). *Let $q \equiv 1, 3, 12, 14 \pmod{16}$ and put $L = B(q)$. Then for some $m \geq 1$ and $\delta \in \{0, 1\}$, the two children of $A(q)$ are exactly*

$$\{Q_m^\delta(J(L)), Q_{m+1}^\delta(J(L))\}.$$

Proof. Write $q = 16u + c$. The four direct formulas are

c	L	$B(A(q))$	$A^2(q)$
1	$R(6u)$	$18u + 1$	$36u + 3$
3	$R(6u + 1)$	$18u + 4$	$36u + 8$
12	$R(6u + 4)$	$18u + 13$	$36u + 27$
14	$R(6u + 5)$	$18u + 16$	$36u + 32$.

For $q = 1$, one has $L = 0$ and the two children of $A(q) = 2$ are $1 = Q_1^1(J(0))$ and $3 = Q_2^1(J(0))$, so the assertion is direct. Otherwise put $z = 6u, 6u + 1, 6u + 4, 6u + 5$ in the four rows. Then $z > 0$, $L = R(z)$, and $B(A(q)) = 3z + 1$. Lemma 2.1 gives $3z + 1 = Q_m^\delta(J(L))$ for some $m \geq 1$, where $\delta = 1 - (z \bmod 2)$. The last column of the table is $2B(A(q)) + \delta$, hence is $Q_{m+1}^\delta(J(L))$. $\square$

Lemma 3.5 (Universal source bound). *For every positive state q ,*

$$\rho(q) \leq \frac{q - 1}{6}.$$

Proof. If $q = Q_r^e(3^k J(s))$, then $r \geq 1$ and $e \leq 1$ give $q \geq 2J(s) - 1$. Since $J(s) \geq 3s + 1$, we have $q \geq 6s + 1$. $\square$

3.1 Two permanent negative regressions

The first invalid shortcut in earlier manuscripts was

$$\text{smaller canonical coefficient} \implies \text{smaller coefficient source.}$$

It is false in the presence of powers of three.

Proposition 3.6 (Coefficient/source separation). *Let $s = 1$, so $J(s) = 5$, and put $a = 3J(s) = 15$, $e = 1$. The common boundary child is*

$$p = R(3a - e) = R(44) = 22.$$

Its canonical odd coefficient is $11 < 15$, but $11 = J(3)$. Hence its source has increased from 1 to 3.

Proof. The displayed arithmetic is direct. In binary, $44 = 101100_2$ and deleting its maximal alternating suffix gives $10110_2 = 22$. Since $22 = Q_1^0(11)$ and $11 = J(3)$, the source is 3. $\square$

The second invalid shortcut compared a return after several expanding moves with the numerical source before the streak.

Proposition 3.7 (Pre-streak anchor separation). *In the conjugated game,*

$$20 \xrightarrow{A} 30 \xrightarrow{A} 45 \xrightarrow{A} 68, \quad B(68) = 25 > 20.$$

Thus a contracting return after a multi-A streak need not lie below the source retained before the streak.

Proof. Each value follows from the definitions of A and B . $\square$

Neither implication is used later in this manuscript.

4 Finite outcomes, witnesses, and retained tokens

Every finite WIN or LOSS state has a canonical proof height. Put $h(0) = 0$ and define

$$h(x) = 1 + \min\{h(y) : y \text{ is a LOSS child of } x\} \quad (x \text{ WIN}),$$

$$h(x) = 1 + \max\{h(y) : y \text{ is a child of } x\} \quad (x \text{ LOSS}).$$

A finite state carried only to justify a local parent/child incidence is a *routing witness*. A *retained proof token* is a finite state whose relation to the existing rank has also been proved: it either uses an explicit one-shot initialization or replaces an already retained token by proper proof-tree descendants. Merely discovering that a state is finite does not make it a retained token.

Lemma 4.1 (Multiset token rank). *For a finite multiset M of proof heights, let*

$$\Phi(M) = \#_{h \in M} \omega^h,$$

where $\#$ is natural ordinal sum. Replacing one occurrence of height H by finitely many certified descendants of heights strictly below H strictly decreases Φ .

Proof. In Cantor normal form, the coefficient of ω^H decreases by one. Every inserted monomial has smaller exponent, so no finite collection of inserted terms can compensate for that loss. $\square$

Remark 4.2 (Why provenance is not cosmetic). A finite WIN state may have another WIN child whose proof height is not controlled by the canonical LOSS branch used to prove the parent WIN. Therefore a later finite state cannot be substituted into an ordinal rank unless it is either initialized by a declared seed or proved to be a proper descendant of a retained token. This is the central audit rule for the remaining sections.

5 A complete reverse fixed-tail progress lemma

The following strengthens the fixed-tail statement used in Versions 5 and 6. The old statement produced a finite witness but did not prove the progress needed after that witness appeared.

Lemma 5.1 (Reverse fixed-tail progress). *Let c be positive and odd, $e \in \{0, 1\}$, and $r \geq 2$. Put*

$$X = Q_r^e(3c), \quad G = Q_r^e(c), \quad H = Q_{r+1}^e(c), \quad Y = A(G).$$

Then

$$\text{Moves}(H) = \{X, Y\}.$$

Let $w = B(X)$. More precisely,

$$w = \begin{cases} B(Y), & r = 2, \\ A(Y), & r \geq 3. \end{cases}$$

If X is DRAW, exactly one of the following progress alternatives is available:

- (i) *H is DRAW; replacing X by H removes one factor of three from the displayed coefficient;*
- (ii) *H is WIN, Y is LOSS, w is a WIN child of Y with $h(w) \leq h(Y) - 1$, and the other child $A(X)$ is DRAW.*

In alternative (ii), the continuation is therefore accompanied by the strict token replacement $Y \mapsto w$ whenever Y has legitimately been initialized as a token.

Proof. Since $r + 1 \geq 3$, Lemma 3.1 gives

$$A(H) = Q_r^e(3c) = X.$$

For $r \geq 3$ it also gives $B(H) = Q_{r-1}^e(3c) = A(G) = Y$; for $r = 2$ the shared boundary identity gives the same equality. Hence $\text{Moves}(H) = \{X, Y\}$.

For $r = 2$, $X = Q_2^e(3c)$ and $Y = Q_1^e(3c)$, so the shared boundary identity gives $B(X) = B(Y)$. If $r \geq 3$, direct use of the long-tail formulas, including the elementary $Q_2 \mapsto Q_1$ row when $r = 3$, gives

$$B(X) = Q_{r-2}^e(9c) = A(Y).$$

This proves the common-child statement.

Assume X is DRAW. Since H has a DRAW child, H is not LOSS. If H is DRAW, alternative (i) holds. If H is WIN, one of its children must be LOSS. Since X is DRAW, that child is necessarily Y , so Y is LOSS. The state w is an actual child of this LOSS state and is therefore WIN, with $h(w) \leq h(Y) - 1$. It is also the B -child of X . A DRAW state has no LOSS child and at least one DRAW child. Since w is WIN, the other child $A(X)$ must be DRAW. This is alternative (ii). $\square$

Corollary 5.2 (Finite reverse factor countdown). *Starting from a displayed DRAW state $Q_r^e(3^k a)$ with $r \geq 2$ and $k > 0$, repeated use of Lemma 5.1 either reaches a factor-free displayed coefficient after finitely many reverse steps, or exposes a strict finite descendant before continuing the DRAW.*

Proof. Every occurrence of alternative (i) reduces k by one. The nonnegative integer k cannot be reduced infinitely. Alternative (ii) is the stated strict finite descendant event. $\square$

Remark 5.3. The replacement $X \rightsquigarrow H$ is a reverse proof normalization, not a legal move from X . Any global use must still preserve an actual continuing DRAW cursor, as required later in Definition 11.1.

6 A complete exponent-one predecessor split

A reverse factor lemma valid for tail exponent at least two must not be used at exponent one. The correct exponent-one predecessor is as follows.

Lemma 6.1 (Exponent-one predecessor progress). *Let a be positive and odd, $k \geq 1$, and put*

$$X = Q_1^e(3^k a), \quad P = Q_2^e(3^{k-1} a).$$

Then $A(P) = X$. If X is DRAW, P is either DRAW or WIN.

If P is DRAW, the displayed factor exponent decreases from k to $k - 1$. If P is WIN, put $L = B(P)$. Then L is LOSS. Moreover:

- (a) *if $P \not\equiv 1, 3, 12, 14 \pmod{16}$, then $w = B(X) = B(A(P))$ is an actual WIN child of L , $h(w) \leq h(L) - 1$, and $A(X)$ is DRAW;*
- (b) *if $P \equiv 1, 3, 12, 14 \pmod{16}$, then the two children of $X = A(P)$ form an exact adjacent factor-free pair*

$$\{Q_m^\delta(J(L)), Q_{m+1}^\delta(J(L))\}$$

for some $m \geq 1$ and $\delta \in \{0, 1\}$, and this pair contains a continuing DRAW.

Proof. The formula $A(P) = X$ is the $Q_2 \mapsto Q_1$ instance of the constant-tail calculation. A state with a DRAW child cannot be LOSS. If P is WIN, the DRAW child X cannot witness the win, so the other child $L = B(P)$ is LOSS.

Assume first that P is ordinary. Then $L > 0$: if $B(P) = 0$, the binary word $A(P)$ is wholly alternating, and the residue argument used in Lemma 3.2 puts P in an exceptional class. Lemma 3.2 therefore gives

$$B(A(P)) \in \{A(B(P)), B(B(P))\} = \text{Moves}(L).$$

Thus $w = B(X) = B(A(P))$ is a child of the LOSS state L , hence is WIN and has height at most $h(L) - 1$. Since X is DRAW and its B -child w is WIN, its other child $A(X)$ is DRAW.

If P is exceptional, write $P = 16t + c$ with $c \in \{1, 3, 12, 14\}$. Lemma 3.4 gives exactly the adjacent pair over $L = B(P)$. Because X is DRAW, neither child is LOSS and at least one is DRAW. $\square$

Remark 6.2. Lemma 6.1 closes the local misuse of reverse factor removal at exponent one. It does *not* by itself prove that the newly exposed L is already a retained token in a global rank. That is a separate initialization/provenance question.

7 The source-zero valuation-one row, corrected

Source zero means that the odd coefficient is a power of three. Put

$$S_{k,r}^e = Q_r^e(3^k).$$

Long tails reduce to $r \in \{1, 2\}$ in finitely many forward moves. The resulting signed boundary has, for some $n \geq 1$,

$$3^n + 1 - 2e = 2^j J(y).$$

Lemma 7.1 (Exact source-zero valuation table). *The valuation is*

	n odd	n even
$e = 0$	$j = 2$	$j = 1$
$e = 1$	$j = 1$	$j = 2 + v_2(n)$.

When $j = 1$,

$$y = \frac{3^{n-1} - 1}{2}.$$

Proof. For $e = 0$, the lifting-the-exponent formula gives $v_2(3^n + 1) = 2$ for odd n , while $3^n + 1 \equiv 2 \pmod{4}$ for even n . For $e = 1$, $3^n - 1$ has valuation 1 for odd n and $2 + v_2(n)$ for even n . It remains to identify y in the two $j = 1$ rows. Put $y = (3^{n-1} - 1)/2$. If $e = 0$, then n is even, so y is odd and

$$J(y) = 3y + 2 = \frac{3^n + 1}{2}.$$

If $e = 1$, then n is odd, so y is even and

$$J(y) = 3y + 1 = \frac{3^n - 1}{2}.$$

These are exactly the odd quotients after the single factor of two is removed. $\square$

Lemma 7.2 (Correct raw selector in the $j = 1$ row). *Assume $j = 1$ and put $d = 1 - e$. The signed exit is*

$$T = Q_1^d(J(y)),$$

and its raw companion is

$$R(T) = \begin{cases} A(y), & e = 0, \\ B(y), & e = 1. \end{cases}$$

Suppose this signed/raw pair is exposed as the two children of a DRAW boundary state, the raw companion is DRAW, and T is finite. Then T is WIN. The following split is exhaustive:

- (i) *if y is WIN, the other ordinary child of y is an exact LOSS sibling of the raw DRAW;*
- (ii) *if y is DRAW, the row yields a reverse DRAW parent y , but it does not yield a finite token or a source decrease.*

Proof. The first identity is the definition of the signed exit. In the $e = 0$ valuation-one row, n is even and $y = (3^{n-1} - 1)/2 \equiv 1 \pmod{4}$. Hence $A(y)$ is even and

$$T = 2J(y) - 1 = 4A(y) + 1$$

is obtained by appending the bits 01 after a final 0; deleting that maximal alternating suffix leaves $A(y)$. In the $e = 1$ row, n is odd and $y \equiv 0 \pmod{4}$, so $A(y)$ is again even and

$$T = 2J(y) = 4A(y) + 2.$$

The appended bits are 10 and the alternating deletion continues through exactly the maximal alternating suffix of $A(y)$, leaving $R(A(y)) = B(y)$.

The raw and signed exits are the two children of the same DRAW boundary configuration. If the signed exit is finite, it cannot be LOSS, because a DRAW state has no LOSS child; hence it is WIN. Since the raw exit is an actual child of y , the state y cannot be LOSS. If y is WIN, its other child must be LOSS because the displayed child is DRAW. If y is DRAW, no finite sibling follows. These are all possible outcomes of y . $\square$

The second branch of Lemma 7.2 was incorrectly compressed in Version 6.3 into a generic “loss-sibling entry.” It is not such an entry. It is one of the explicit global obligations below.

8 High returns: the identities are proved, the rank provenance is not

The long high-return coordinates used in Version 6.3 are arithmetically correct. Let $t > 0$, $g \in \{0, 1\}$, and $v \geq 7$. Put

$$u = Q_{v-1}^g(J(t)), \quad c = Q_{v-3}^g(3J(t)), \quad p = B(A(u)), \quad b = B(p).$$

Lemma 8.1 (Exact long high-return diamond). *The identities*

$$p = A(c) = Q_{v-4}^g(9J(t)), \quad b = Q_{v-6}^g(27J(t))$$

hold. Suppose, in addition, that the outcome guard proves

$$u, c, p, b \text{ are WIN}, \quad A(u) \text{ is LOSS}, \quad B(c) \text{ is LOSS}.$$

Then

$$h(p) \leq h(u) - 2, \quad h(b) \leq h(c) - 2.$$

Consequently, if u, c are already retained tokens, the replacement

$$(u, c) \mapsto (p, b)$$

is strict in the multiset token rank. A declared one-shot seed may first install the pair $\{u, c\}$; after that installation the same replacement is strict, while the initialization itself must be paid by the separate seed component rather than by Φ .

Proof. Constant-tail reduction gives

$$A(u) = Q_{v-2}^g(3J(t)), \quad B(A(u)) = Q_{v-4}^g(9J(t)) = A(c),$$

where the boundary rows $v = 7$ are included by direct substitution. Applying the same calculation to p gives the formula for b .

Here $B(u) = c$, and the outcome guard says that c is WIN. Thus $A(u)$ is the unique LOSS child of the WIN state u . Since p is a child of $A(u)$,

$$h(p) \leq h(A(u)) - 1 = h(u) - 2.$$

Similarly, $v - 3 \geq 4$ makes $c \equiv 0$ or $15 \pmod{16}$, so c is ordinary. Its child $A(c) = p$ is WIN, while the guard says that $B(c)$ is LOSS; hence $B(c)$ is the unique LOSS child witnessing c as WIN. Lemma 3.2 makes $b = B(A(c))$ a child of $B(c)$, and therefore

$$h(b) \leq h(B(c)) - 1 = h(c) - 2.$$

Lemma 4.1 gives the final conditional conclusion. □

Remark 8.2 (The precise old gap). The hypotheses “ u and c are finite WIN states” do not imply that they are already members of a ranked multiset. Version 6.3 silently made that promotion. Lemma 8.1 deliberately separates the valid height inequalities from the still missing global initialization and continuity invariant.

9 The two shortest marked tails

The shortest marked-tail arithmetic can also be stated without overclaim. Let C be positive and odd and put $b = Q_D^g(C)$.

Lemma 9.1 (The $D = 1$ row pays an actual loss token). *Assume*

$$q = B(b) \text{ is WIN,} \quad y = A(b) \text{ is LOSS,} \quad D = 1.$$

Factor

$$9J(b) + 1 - 2g = 2^j J(w).$$

Then $j \geq 2$ and

$$w = \begin{cases} A(y), & j = 2, \\ B(y), & j \geq 3. \end{cases}$$

Thus w is an actual WIN child of the retained LOSS token y and $h(w) \leq h(y) - 1$.

Proof. Since $b = 2C - g$,

$$9J(b) + 1 - 2g = \begin{cases} 2(27C + 5), & g = 0, \\ 2(27C - 5), & g = 1. \end{cases}$$

The parenthesized integer is even because C is odd, so $j \geq 2$. With $y = A(b) = 3C - g$, the same identity is

$$9J(b) + 1 - 2g = 2(3J(y) + 1 - 2(1 - g)).$$

The expression in parentheses is the signed source transition at y in phase $1 - g$. Lemma 3.3 says that valuation one in that inner transition (total $j = 2$) selects $A(y)$, while larger valuation selects $B(y)$. Both are children of the LOSS state y . $\square$

Lemma 9.2 (Exact $D = 2$ lift). *Assume*

$$q = A(b) \text{ is WIN,} \quad y = B(b) \text{ is LOSS,} \quad D = 2.$$

Factor

$$9J(b) + 1 - 2g = 2^j J(w).$$

Then $j = 1$. If $y > 0$, the returned source is

$$w = Q_m^{1-g}(J(y))$$

for an exact integer $m \geq 2$. If $y = 0$, the returned state has coefficient source zero.

Proof. Now $b = 4C - g$, and

$$9J(b) + 1 - 2g = \begin{cases} 2(54C + 5), & g = 0, \\ 2(54C - 5), & g = 1. \end{cases}$$

The parenthesized integer is odd, so the valuation is exactly one. Put $z = 3C - g$. The shared exponent-two/one boundary gives $y = R(z)$. When $g = 0$, the integer z is odd and Lemma 2.1 gives

$$3z + 1 = 2^\lambda J(y).$$

The returned source is $w = 18C + 1 = 2(3z + 1) - 1$, and therefore

$$w = Q_{\lambda+1}^1(J(y)).$$

When $g = 1$, the integer z is even and the same lemma gives

$$3z + 2 = 2^\lambda J(y).$$

Now $w = 18C - 2 = 2(3z + 2)$, so

$$w = Q_{\lambda+1}^0(J(y)).$$

Here $\lambda \geq 1$ is the maximal alternating-suffix length of z when $y > 0$. Thus in both phases $w = Q_{\lambda+1}^{1-g}(J(y))$ and $m = \lambda + 1 \geq 2$. If $R(z) = 0$, Lemma 2.1 instead gives a constant-tail state over $J(0) = 1$, which is exactly the source-zero alternative. $\square$

Lemma 9.2 proves attachment of the first returned cursor to the actual LOSS token y . It does not prove that every later raw and signed exit preserves that attachment until a descendant is paid. That latter statement is the second principal global obligation.

10 The exact remaining global obligations

The earlier versions attempted to assemble local arithmetic into a marked transition system with source anchors, one-shot token seeds, proof-token multisets, and finite control counters. The order-theoretic idea is sound, but the semantic refinement to every legal DRAW continuation is not yet complete. The missing content is now isolated as follows.

Open obligation 10.1 (Z: source-zero $j = 1$ bootstrap). In the second branch of Lemma 7.2, where both the raw child and its ordinary parent y are DRAW, construct a marked transition that either pays a previously retained source/token component or consumes a declared one-shot bootstrap which cannot be consumed again in the same earlier fibre. Prove that every later return to source zero is compatible with this non-reseeding rule.

Open obligation 10.2 (H: high-return pair provenance). Reserve an inner one-shot seed for the first long high-return pair $\{u, c\}$, or prove that the pair consists of descendants of already retained tokens. After the seed has been consumed, prove for every subsequent long high return in the same outer fibre that its incoming pair is exactly the currently retained pair or certified descendants of it. No merely finite routing witness may be promoted to the rank.

Open obligation 10.3 (L: complete LOSS-anchored $D = 2$ module). Starting from the exact lift of Lemma 9.2, enumerate every legal raw/signed outcome branch, every remaining factor exponent, and the terminal tail classes $m = 1, 2, 3, \geq 4$. Prove that each macro-step either decreases an explicit finite counter while retaining the same LOSS token y , or replaces y by an actual child or certified descendant before any generic obligation is re-entered. In particular, prove that no branch exits as a fresh free obligation at an unrelated arithmetic cursor.

Open obligation 10.4 (E: semantic exhaustiveness of the combined router). After Obligations 10.1–10.3 are supplied, construct a marked macro-relation satisfying every clause of Definition 11.1 below. In particular, give a disjoint and complete case partition for every control transition. For each equal-rank cycle, exhibit a natural counter that strictly decreases; for each control reset, exhibit a preceding strict source or token replacement. This proof must refine the actual game outcomes, not only an acyclic graph of declared control labels.

Remark 10.5. Lemma 5.1 closes the former fixed-tail part of the entry countdown, and Lemma 6.1 closes the local exponent-one misuse. They reduce Obligation 10.4, but do not imply Obligations 10.1–10.3.

11 A complete conditional closure theorem

This section records the exact abstract statement supplied by the marked-rank architecture. The theorem is conditional, but the condition is now a mathematically explicit property rather than the name of a control graph.

Let a marked configuration retain

$$\Pi = (s, \eta, \Phi(M), \zeta, \Psi(N), a),$$

ordered lexicographically. Here $s, a \in \mathbb{N}_0$ are genuine retained source anchors, M, N are finite multisets of certified token heights, $\Psi(N)$ is defined by the same natural-sum formula as $\Phi(M)$, and $\eta, \zeta \in \{1, 0\}$ are one-shot seed bits ordered by $1 > 0$. With the usual orders on the natural and ordinal components, this finite lexicographic product is a well-order. Temporary sources, phases, valuations, and tail lengths are not components of Π unless an explicit strict comparison installs them as retained data.

Definition 11.1 (Complete marked-router hypothesis). A marked macro-relation $\implies$ is called complete for this game when all of the following hold.

- (C1) Every conjugated DRAW state produces an initial marked configuration carrying that same actual DRAW as its continuing cursor.
- (C2) From every marked configuration carrying an actual DRAW, a finite outcome-compatible normalization produces a successor marked configuration carrying an actual DRAW. Reverse-parent transformations are permitted only as proof transformations; whenever the cursor follows play, it follows a legal DRAW child.
- (C3) Every macro-step weakly decreases Π . A token-changing step is a one-shot seed or a certified proper-descendant replacement; a source-changing step uses an explicit strict comparison of retained sources.
- (C4) For each fixed value of Π , the restriction of $\implies$ to that fibre is well-founded. In particular, every equal- Π control cycle has an explicitly decreasing natural counter.

Lemma 11.2 (Well-founded fibre principle). *Let a transition relation carry a value in a well-order. Suppose every transition weakly decreases that value and the restriction of the relation to each fixed-value fibre is well-founded. Then the full relation is well-founded.*

Proof. An infinite path would induce a nonincreasing sequence in a well-order. If it had infinitely many strict decreases, those decreases would form an infinite descending subsequence, which is impossible. Hence its tail would lie in one fixed fibre, contradicting fibre well-foundedness. $\square$

Theorem 11.3 (Conditional no-draw closure). *If a complete marked macro-relation in the sense of Definition 11.1 is constructed from the game relation, then the conjugated game has no DRAW positions. Consequently every original positive odd start has finite remoteness.*

Proof. Assume a conjugated DRAW exists. Clause (C1) gives an initial marked configuration carrying it. Clause (C2), applied recursively, gives an infinite sequence of marked configurations, each carrying an actual continuing DRAW. Clauses (C3)–(C4) and Lemma 11.2 say that the same macro-relation is well-founded, so such an infinite sequence cannot exist. Hence no conjugated DRAW exists. Proposition 2.2 transfers the conclusion to every original odd start: after the first move both possible children are conjugated finite-remoteness states, so the original state is itself WIN or LOSS with finite height. $\square$

Remark 11.4. Theorem 11.3 is not a proof of Conjecture 1.2. Obligations 10.1–10.3 are specific missing semantic modules, and Obligation 10.4 is the remaining task of constructing a relation satisfying Definition 11.1. None is assumed silently in an unconditional theorem.

12 Computational verification boundary

The companion program `verify_v7_0.py` checks, over configurable finite ranges:

- (1) the original-to-binary normal form and strict contraction of B ;
- (2) the long-tail recurrence and shared boundary;
- (3) the alternating-suffix factorization, the bijection J , ordinary coefficient-source recovery, the ordinary side relation, the source-boundary selector, and the exceptional child-pair identity;
- (4) the two permanent negative regressions;
- (5) every arithmetic identity in Lemmas 5.1, 6.1, 7.1, 7.2, 8.1, 9.1, and 9.2.

The program does not infer infinite outcome classes and does not check Obligations 10.1–10.4. Its final message explicitly says that the run is supporting evidence, not a no-draw proof.

13 Hostile-audit matrix

Item	Status in v7.0	Audit conclusion
Binary conjugation and $B(q) < q$	Proved	Exact algebra; finite regression included.
Long-tail identities	Proved	Uniform in a, r, e ; no bounded-modulus assumption.
Coefficient versus source	Corrected	The explicit $1 \rightarrow 3$ counterexample is retained permanently.
Multi- A numerical anchor	Corrected	The $20 \rightarrow 30 \rightarrow 45 \rightarrow 68$, $B(68) = 25$ counterexample is retained permanently.
Fixed-tail reverse entry	Proved locally	Lemma 5.1 adds the missing continuing-DRAW and descendant-token alternatives.
Exponent-one reverse entry	Proved locally	Lemma 6.1; no use of the $r \geq 2$ reverse lemma at $r = 1$.
Source-zero $j = 1$ row	Partly proved	Arithmetic and outcome split proved; global non-reseeding bootstrap is Obligation 10.1.
Long high-return heights	Proved conditionally on token legitimacy	The inequalities are valid; rank insertion is isolated as Obligation 10.2.
Marked tail $D = 1$	Proved	Actual child of the retained LOSS token.
Marked tail $D = 2$	Exact first lift proved	Full raw/signed provenance remains Obligation 10.3.
Complete router	Open for the target theorem	No unconditional exhaustiveness claim; Obligation 10.4.
No-draw conclusion	Conditional only	Theorem 11.3 under Definition 11.1; Conjecture 1.2 remains open here.

14 Conclusion

Version 7.0 does not repeat the cycle in which one local repair is followed by a new hidden global assumption. It makes three changes of principle. First, it proves complete local progress statements for the fixed-tail and exponent-one reverse rows rather than merely naming a finite witness. Second, it never identifies a routing witness with a retained token without a seed or a descendant comparison. Third, it records the prize statement as a conjecture until the source-zero bootstrap, high-return pair continuity, LOSS-anchored $D = 2$ module, and combined semantic exhaustiveness have all been proved.

The result is a manuscript whose stated theorems are auditable and whose remaining barrier is explicit. It is not yet a solution of the prize problem. Any future Version 8 claiming the no-draw theorem must contain proofs of Obligations 10.1–10.4, or replace the marked-router architecture by a genuinely different complete argument.

Acknowledgment of the audit

Questions from Ingo Althoefer and subsequent hostile review exposed the coefficient/source error, the exponent-one misuse, and the later token provenance failures. Version 7.0 treats those findings as permanent regressions rather than cosmetic edits.

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